

From Dust to Cosmology: Hierarchical Inference using BayeSN

Matt Grayling
Institute of Astronomy

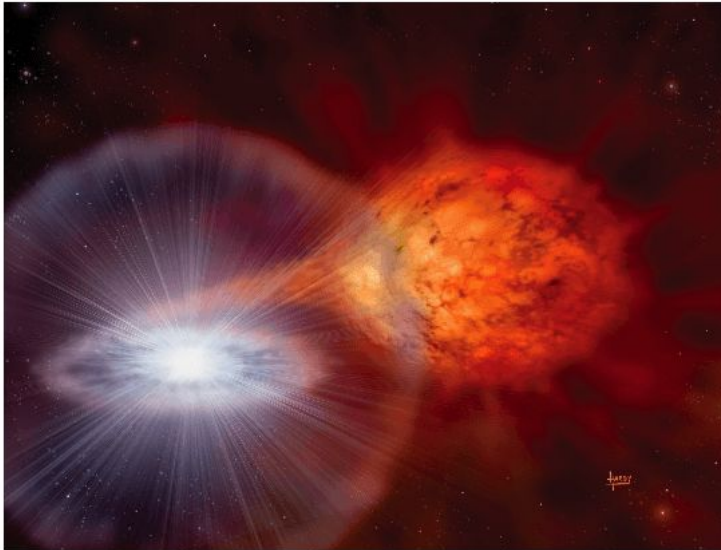


- 1) Unknowns with type Ia supernovae
- 2) Investigating the 'mass step' with BayeSN
- 3) Simulation-based inference - the future of supernova cosmology?

Unknowns in Type Ia Supernovae

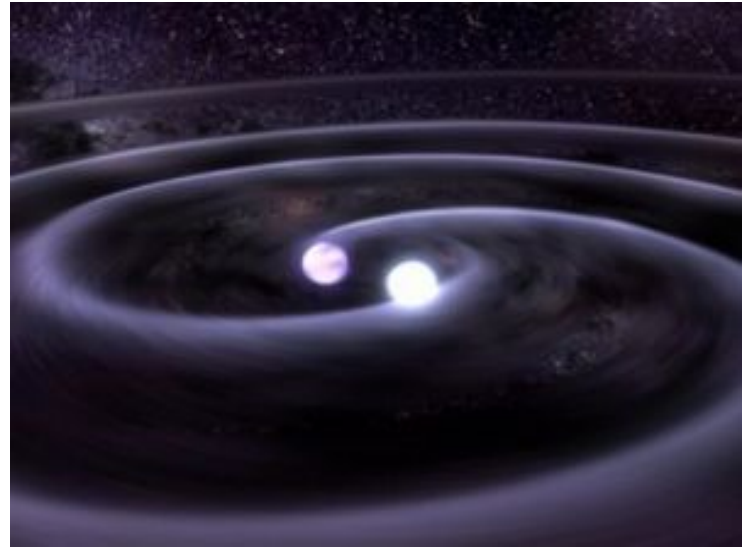
Type Ia Supernovae

Single degenerate



Credit: David A. Hardy, © David A. Hardy/astroasrt.org

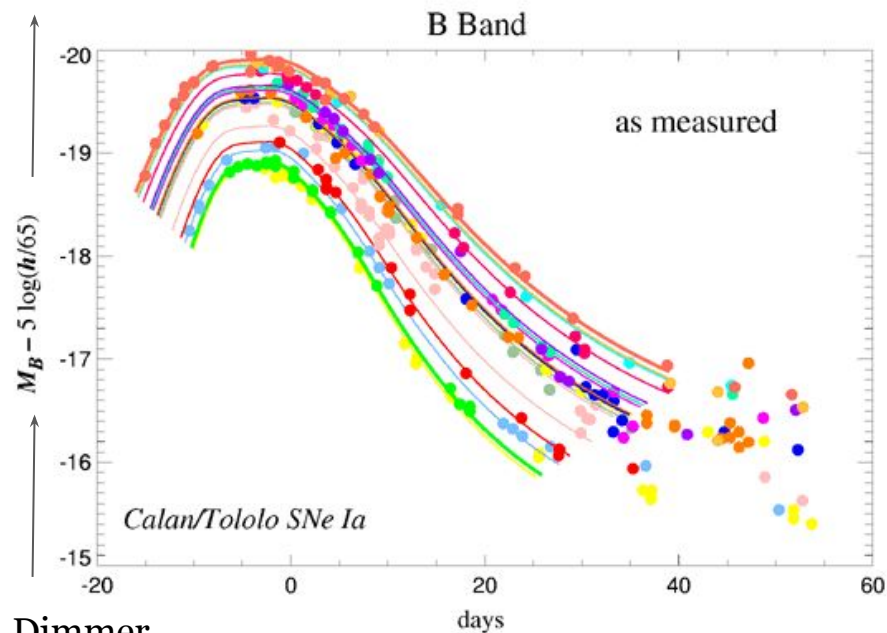
Double degenerate



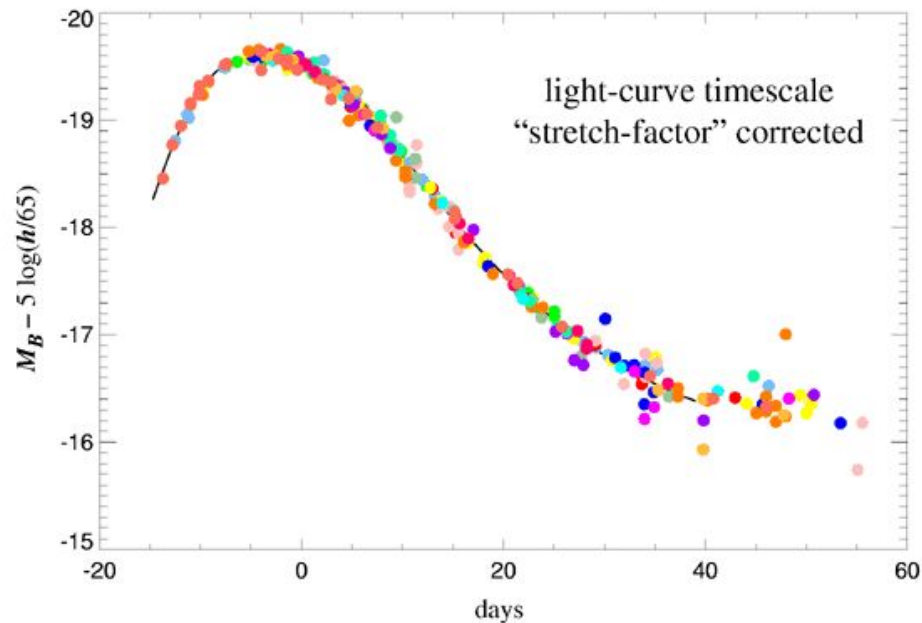
Copyright: GSFC/D. Berry.

Standard(isable) Candles

Brighter

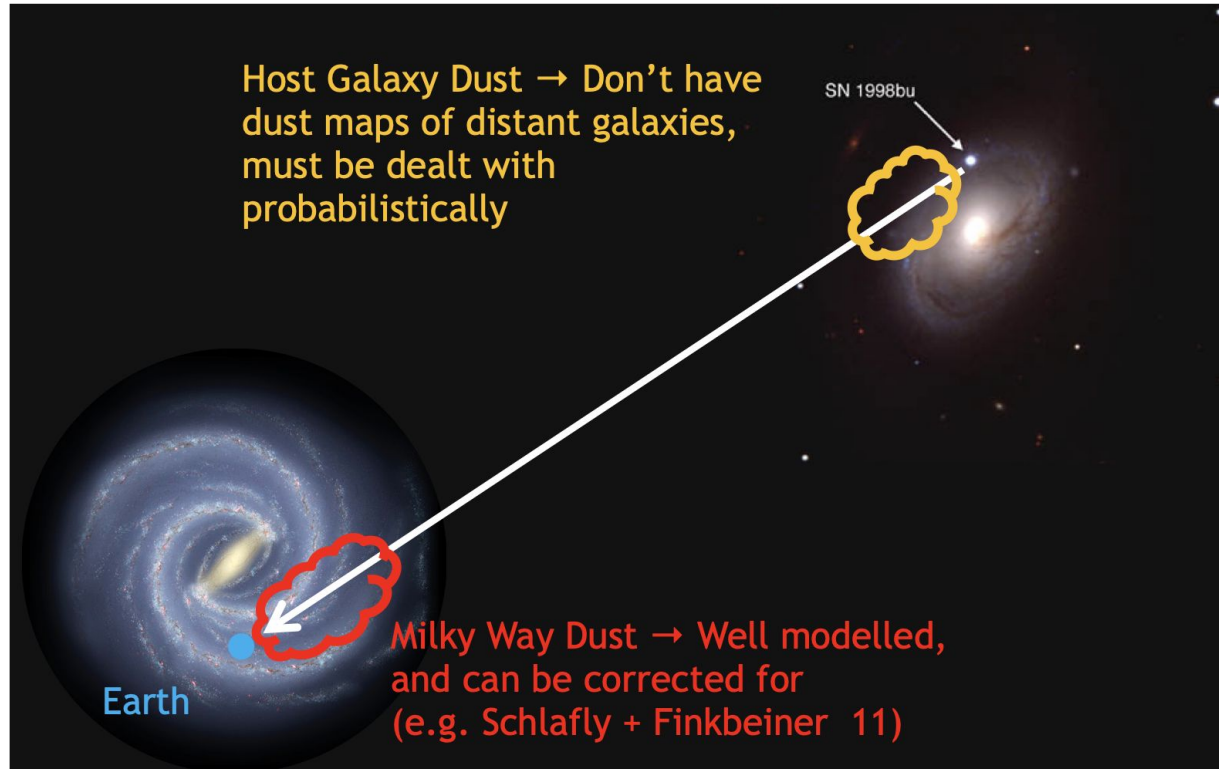


Dimmer



From Supernova Cosmology Project

$$\mu^s = m_B^s - M + \alpha x_1^s + \beta c^s$$



NASA/JPL-Caltech/ESO/R. Hurt
Nicholas B. Suntzeff

$$\mu^s = m_B^s - M + \alpha x_1^s + \beta c^s$$



One parameter, two effects

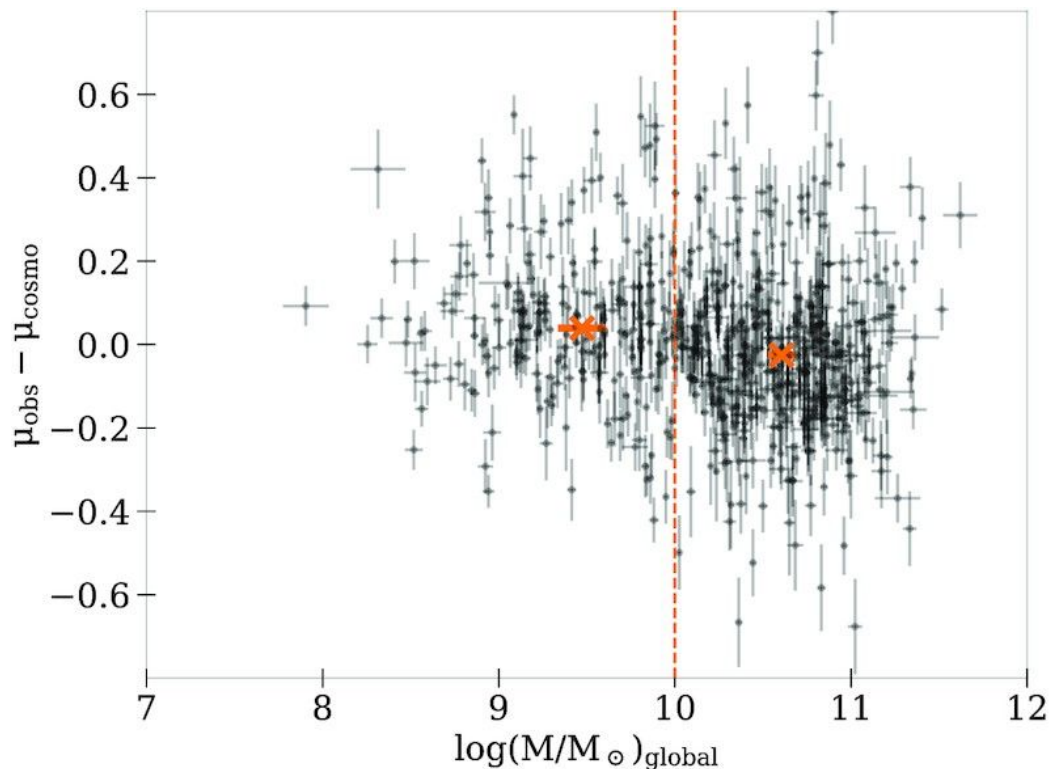
The Significance of Dust

Accurate treatment of dust is key for:

- Understanding the intrinsic properties of type Ia supernovae
- Avoiding bias in distance measurements

The 'Mass Step'

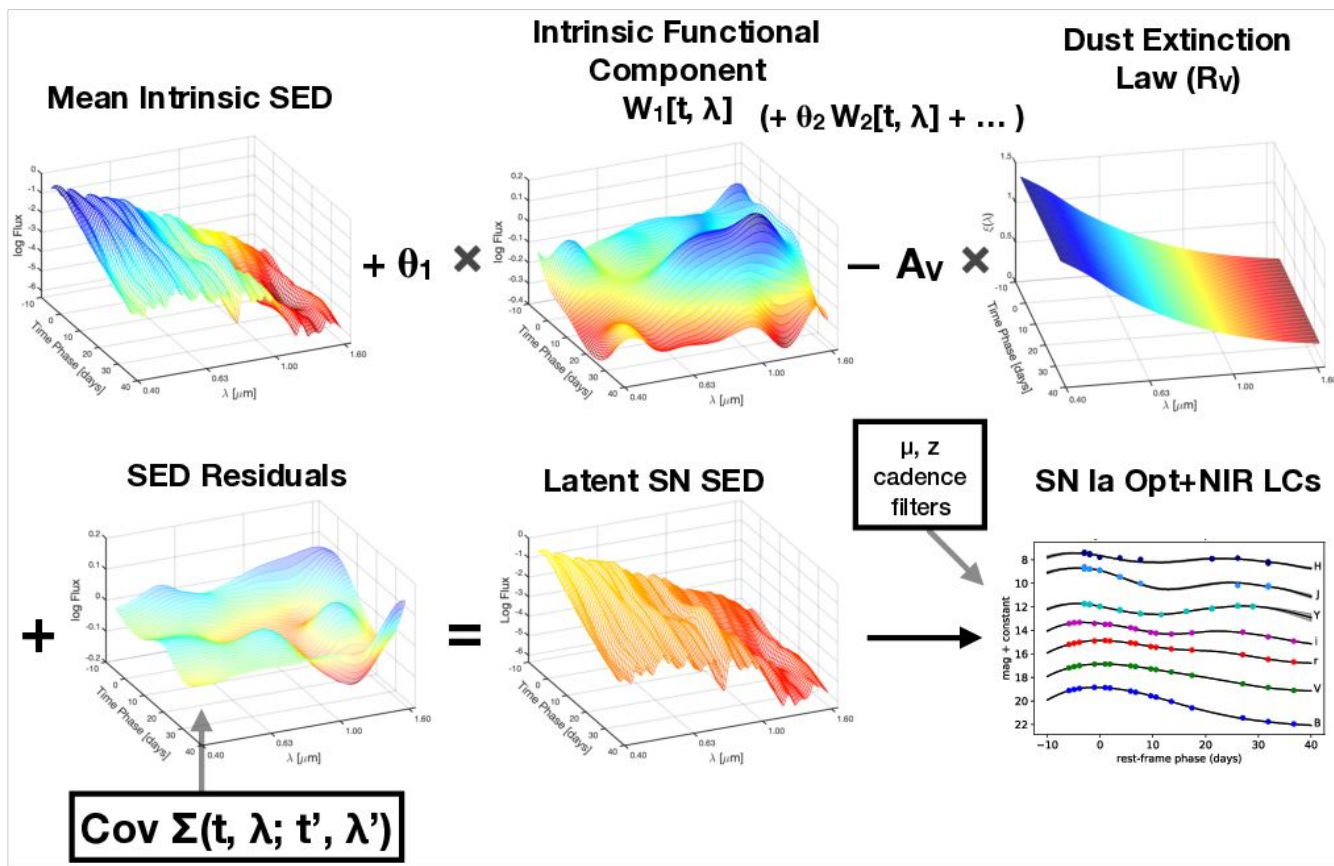
- Disentangling intrinsic variation from dust vital to understanding host galaxy 'mass step'
- Just different dust properties?
 - SN populations identical, just appear different?
- Or intrinsic differences?
 - Interesting physical implications, different progenitor channels?



- Stellar mass can only be a proxy for something else
- Other relations have been observed e.g. with U-R colour, star formation rate
- These relations are stronger when considering local properties
 - Derived from the position around the supernova, rather than the whole galaxy
- Maybe these are a proxy for age
 - Two populations of type Ia supernovae, one from young progenitors and one from old?

Investigating the Mass Step with BayeSN

The BayeSN Model



- For typical Bayesian inference, each supernova would be fit separately

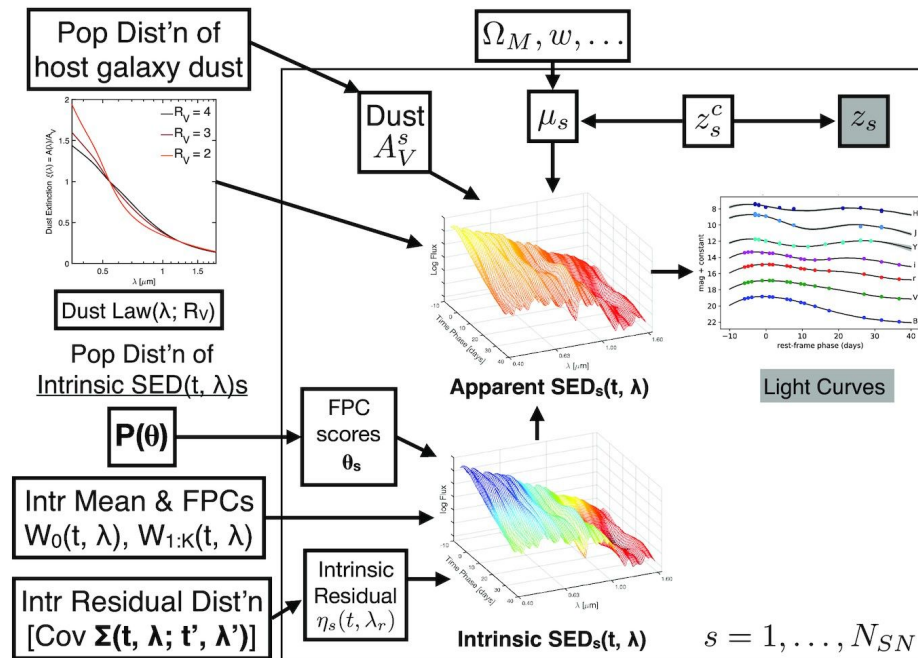
$$R_V \sim U(1, 5)$$

- For a hierarchical model, we jointly sample the posteriors of global and individual supernova properties

$$R_V \sim N(\mu_R, \sigma_R^2)$$

- We can model intrinsic variation and dust properties at the population level as two separate effects

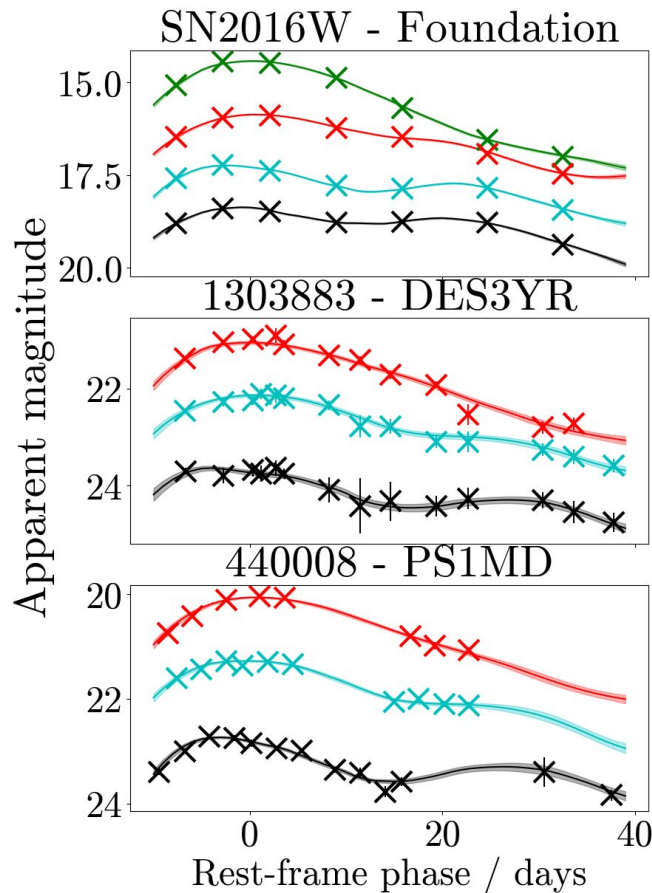
- Model extends into NIR wavelengths
 - Better SN Ia standardisation and constraint of host galaxy dust



Mandel+2020

Investigating the Mass Step

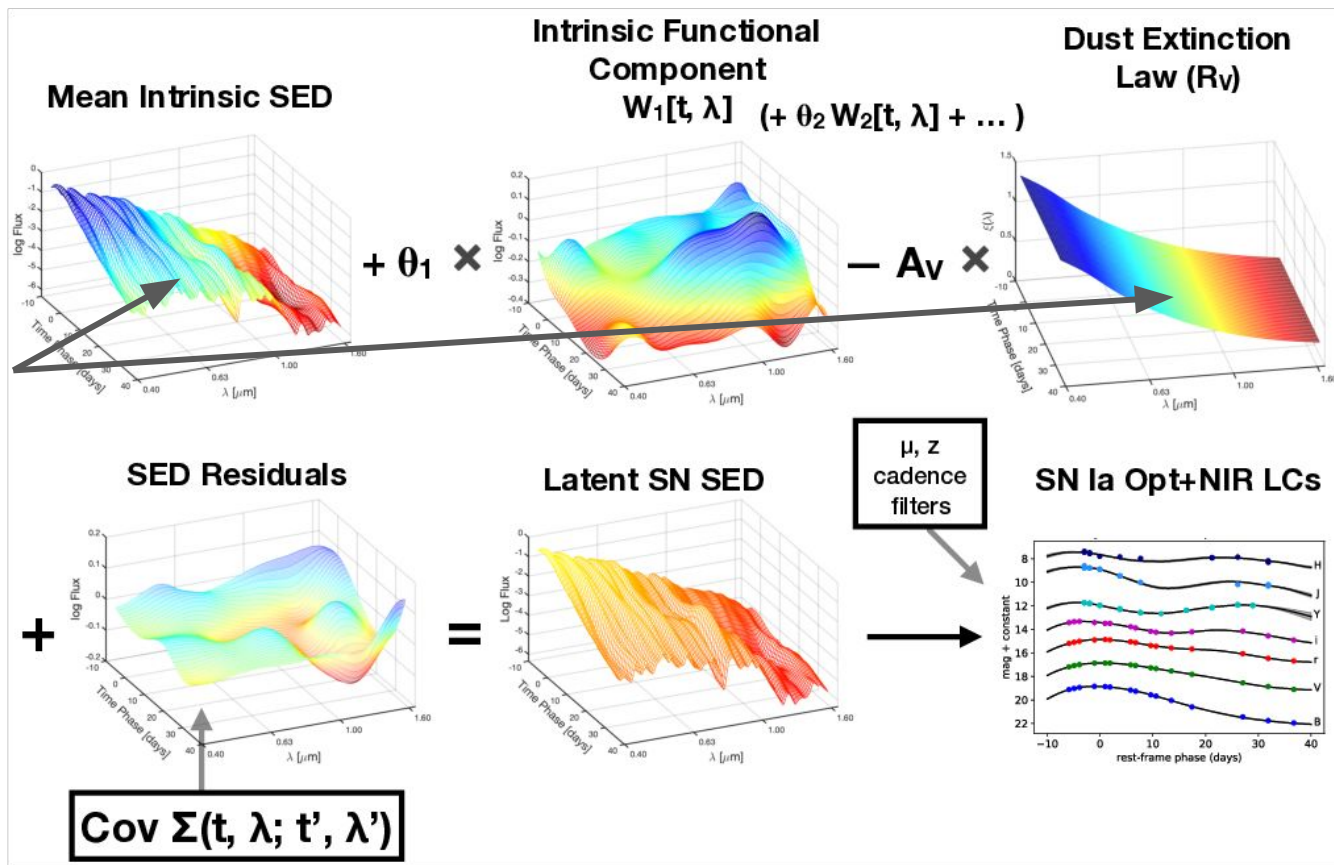
- Infer properties of SN populations separately on each side of the mass step using BayeSN
- Combined sample of 475 SNe Ia from Foundation, DES3YR and PS1MD
- Jointly infer dust and intrinsic differences, see which is more consistent with data



Example
BayeSN
fits

The BayeSN Model

Allow to
differ either
side of
mass step

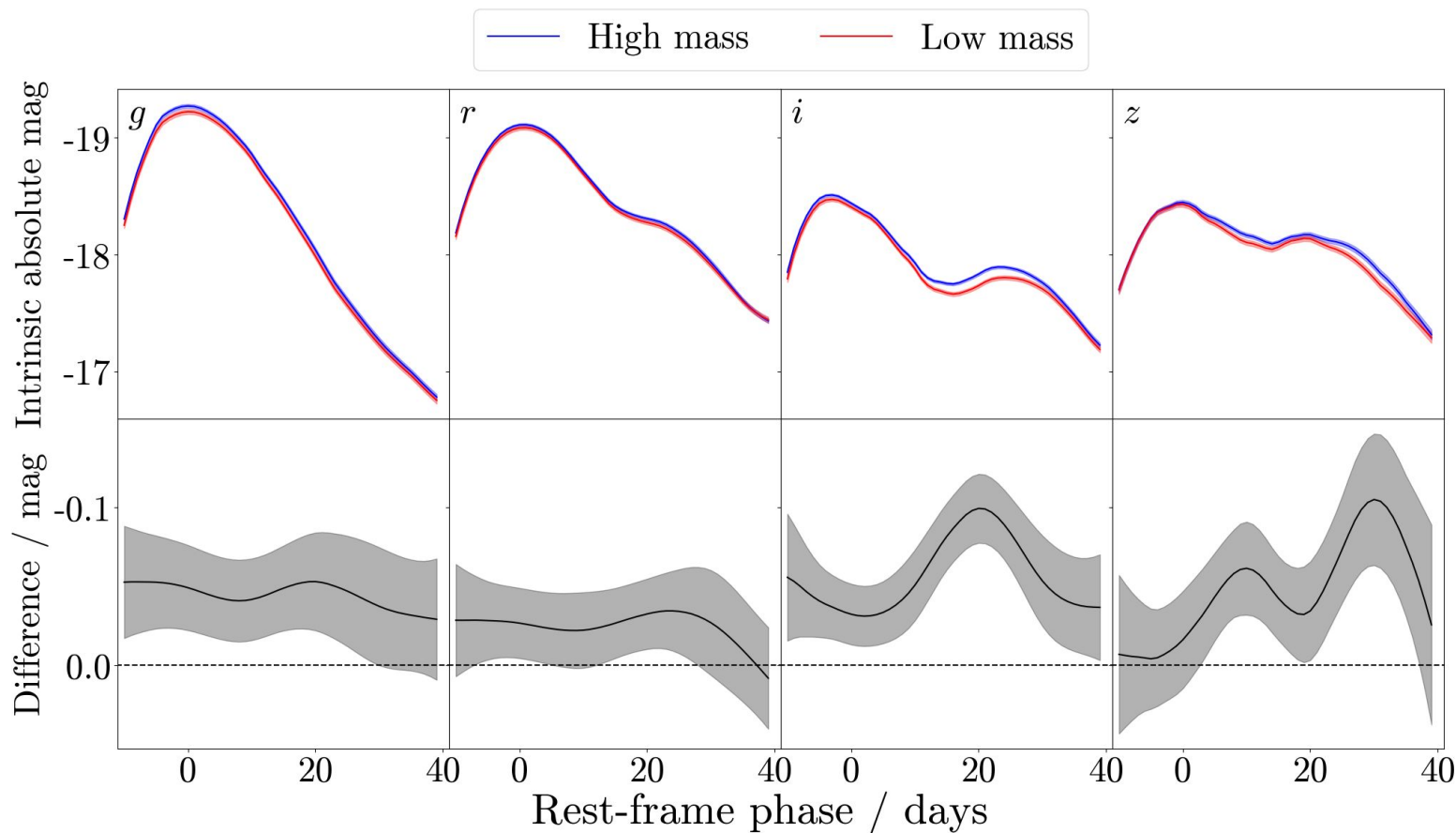


Mandel (2020)

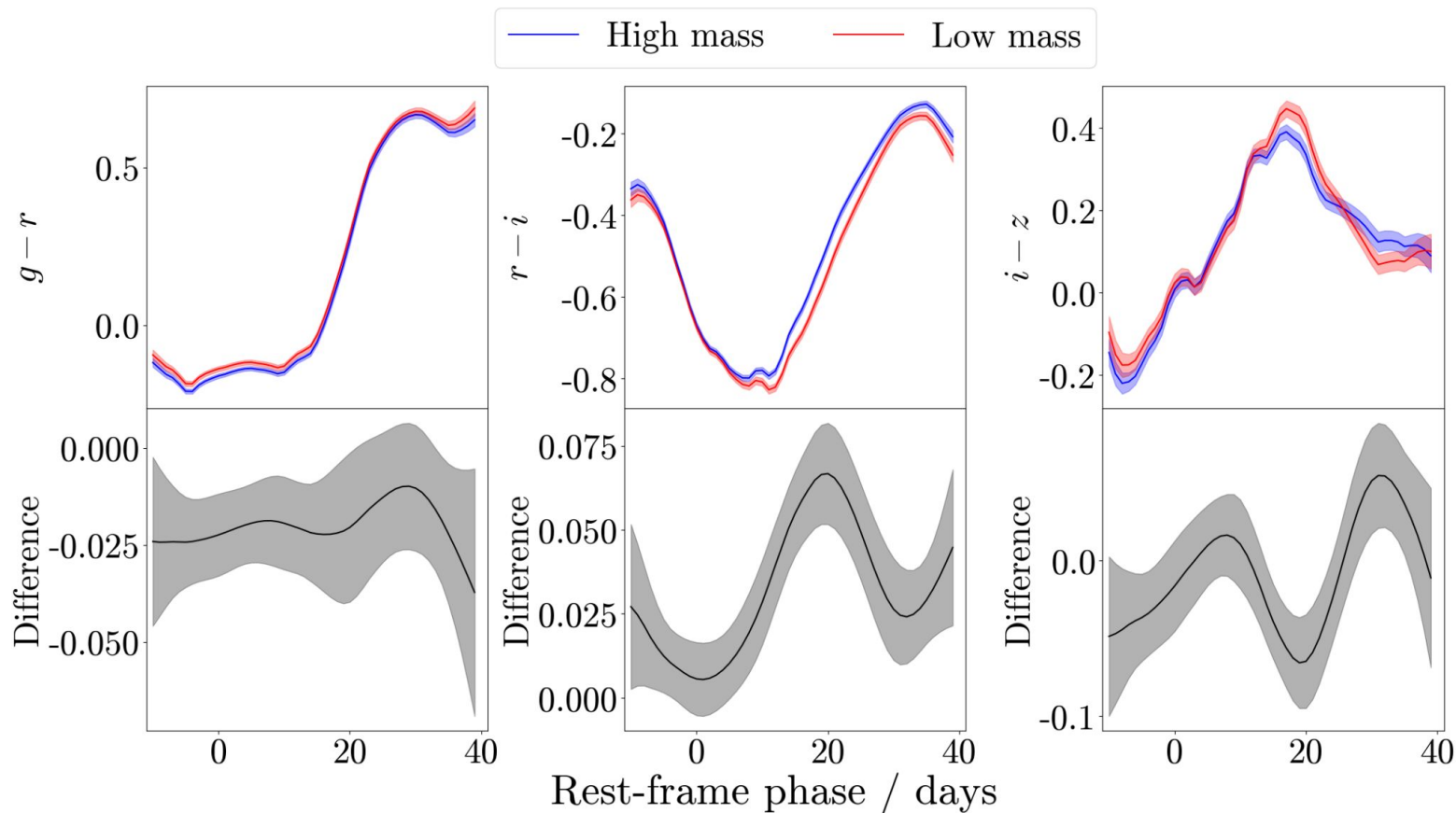
Method	ΔM_0 / mag	$\Delta(g - r)_{\text{peak}}^{\text{int}}$	μ_R		τ_A / mag	
			HM	LM	HM	LM
Intrinsic mag diff.	-0.049 ± 0.016	—	2.51 ± 0.16	2.74 ± 0.35	0.18 ± 0.01	0.11 ± 0.01
Intrinsic SED diff.	-0.049 ± 0.027	-0.022 ± 0.010	2.26 ± 0.14	3.36 ± 0.51	0.19 ± 0.02	0.15 ± 0.02

- Our results support the existence of intrinsic differences between SNe Ia in each mass bin
- Depending on our assumptions, our results also indicate differences in host galaxy dust properties

Intrinsic Baseline Light Curves



Intrinsic Baseline Colour Curves



Method	ΔM_0 / mag	$\Delta(g - r)_{\text{peak}}^{\text{int}}$	μ_R		τ_A / mag	
			HM	LM	HM	LM
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- Our results support the existence of intrinsic differences between SNe Ia in each mass bin
- Depending on our assumptions, our results also indicate differences in host galaxy dust properties
- Lots of different, SED-level effects. Analyses of larger samples will be able to further constrain the relative balance of intrinsic and extrinsic effects

BayeSN Out Now!

New GPU-accelerated (jax/numpyro), Python implementation of BayeSN out now!

[arXiv:2401.08755](https://arxiv.org/abs/2401.08755), github.com/bayesn/bayesn, pip installable, SNANA integration in progress

 README  MIT license

BayeSN

Hierarchical Bayesian SED modelling of type Ia supernova light curves

Developed and maintained by: Matt Grayling (@mattgrayling), Stephen Thorp (@stevet40), Gautham Narayan (@gnarayan), and Kaisey S. Mandel (@CambridgeAstroStat) on behalf of the BayeSN Team (@bayesn).

About

BayeSN is a probabilistic optical-NIR SED model for type Ia supernovae, allowing for hierarchical analysis of the population distribution of physical properties as well as cosmology-independent distance estimation for individual SNe. This repository contains an implementation of the BayeSN SED model built with numpyro and jax, with support for GPU acceleration, as discussed in Grayling+2024 (submitted to MNRAS).

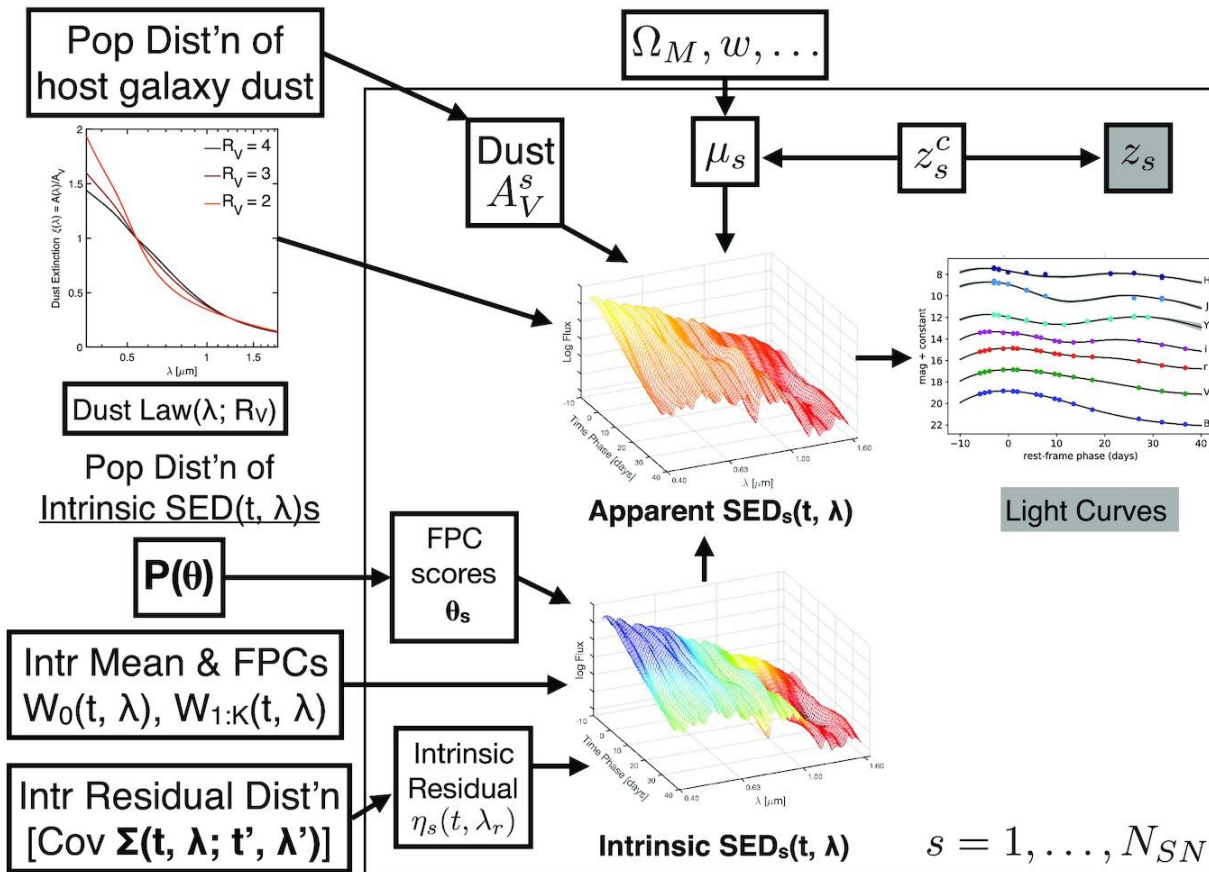


Scan for arXiv link

- Exciting applications to new data sets (DES5YR, ZTF, CSP etc.)
- Training on spectra
- Integration within SNANA
- Fully hierarchical cosmology

Simulation-based Inference: The Future of Supernova Cosmology?

A fully hierarchical cosmology?



$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Diagram illustrating the components of Bayes' Theorem:

- Posterior**: $P(\theta|D)$
- Likelihood**: $P(D|\theta)$
- Prior**: $P(\theta)$
- Evidence**: $P(D)$

The Problem

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Likelihood

$$\mathcal{L} = \frac{1}{\sqrt{2\pi\sigma_f^2}} \exp \left[-\frac{(f_m - \hat{f})^2}{2\sigma_f^2} \right]$$

Gaussian likelihood - easy to calculate

$\mathcal{L} = ?$

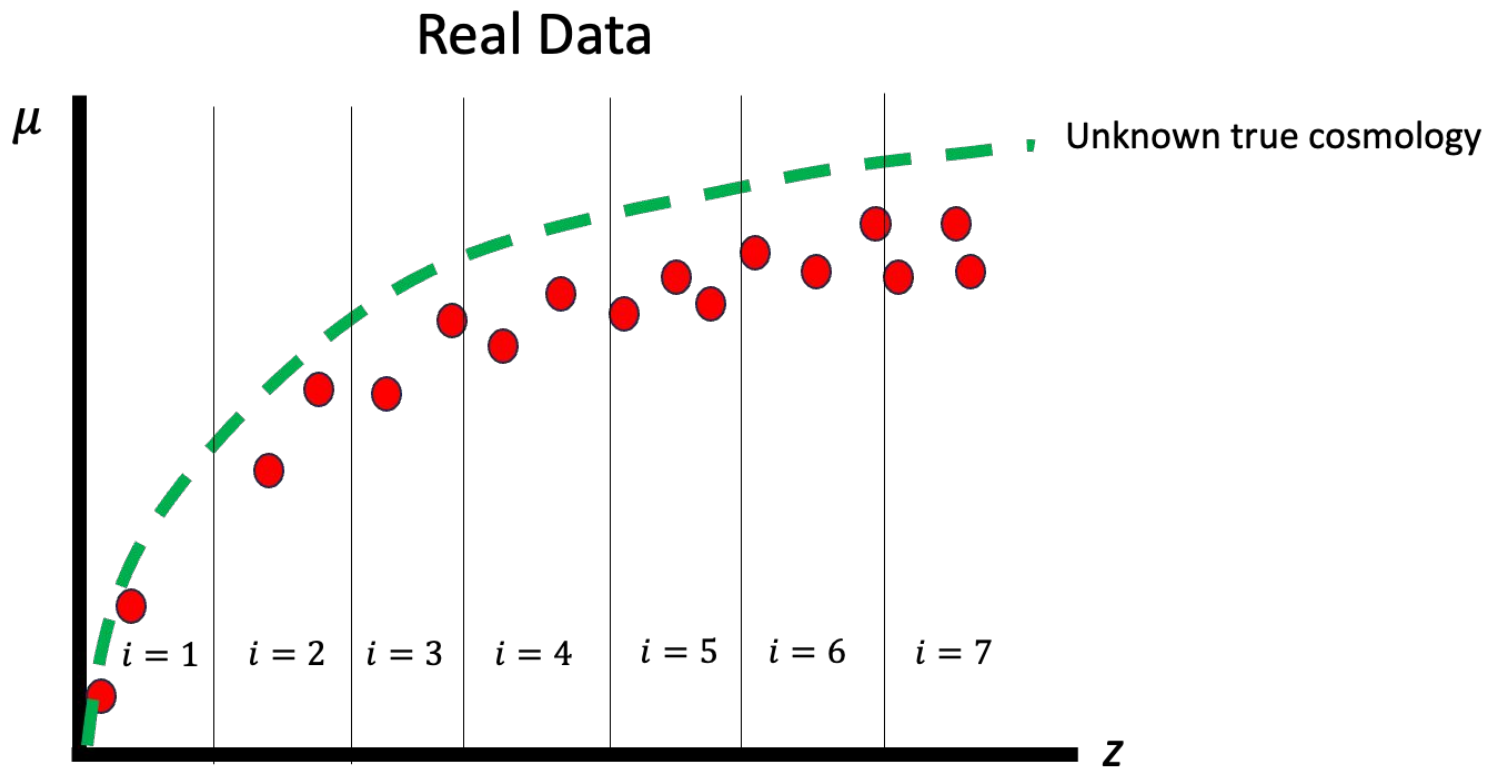
Sometimes the likelihood is too complex to write down

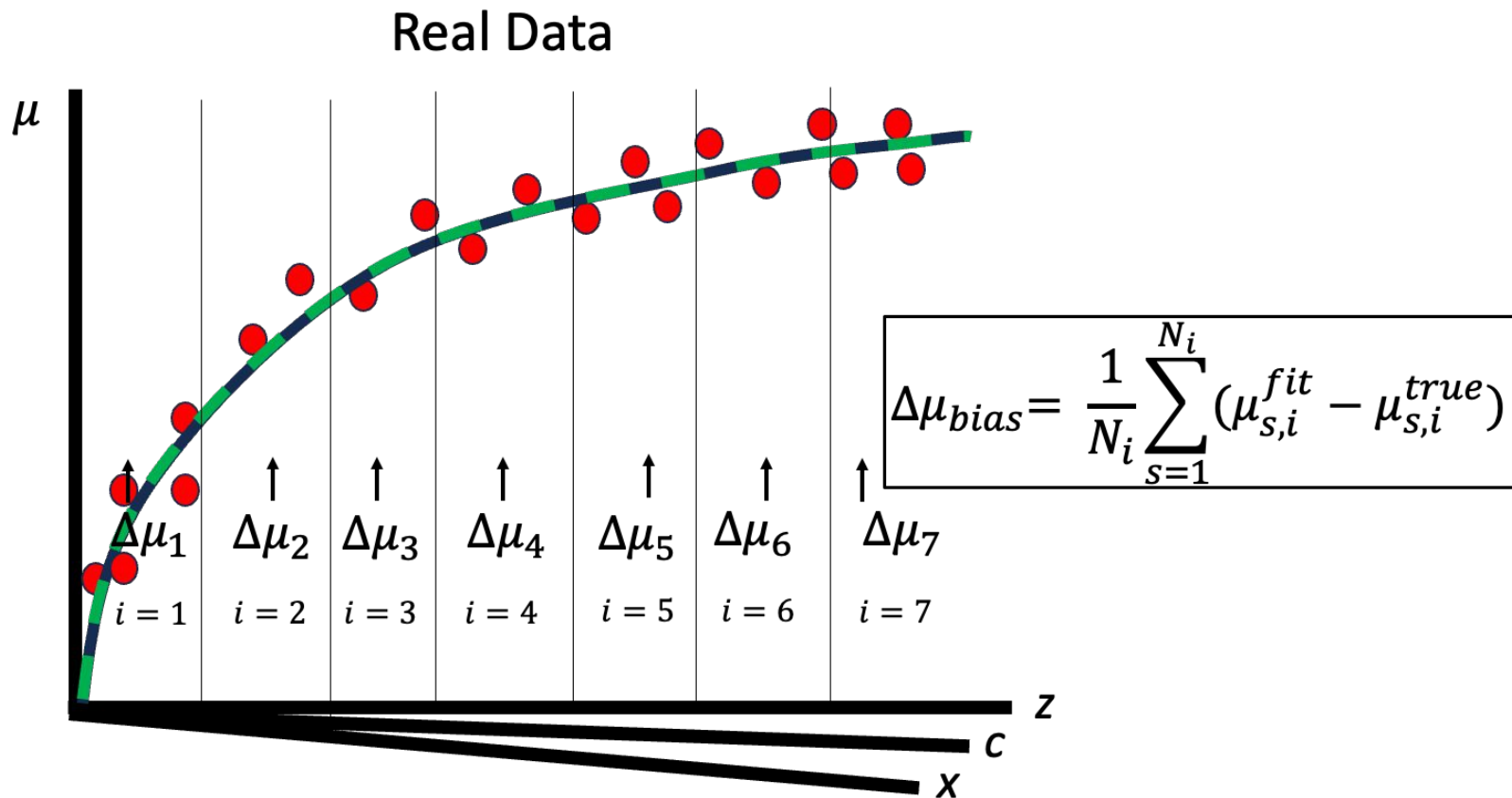
Examples of selection criteria

- More than N observations with signal-to-noise $> 3/4/5$
- More than M observations before/after peak
- $|x_1| < 3$, $|c| < 0.3$
- Need a spectroscopic host redshift

We have developed tools to simulate supernova surveys well, but expressing their selection effects analytically is impossible

$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s + \gamma + \Delta\mu_{\text{bias}}$$





What if we could incorporate selection effects within the hierarchical model, without ever calculating a likelihood?

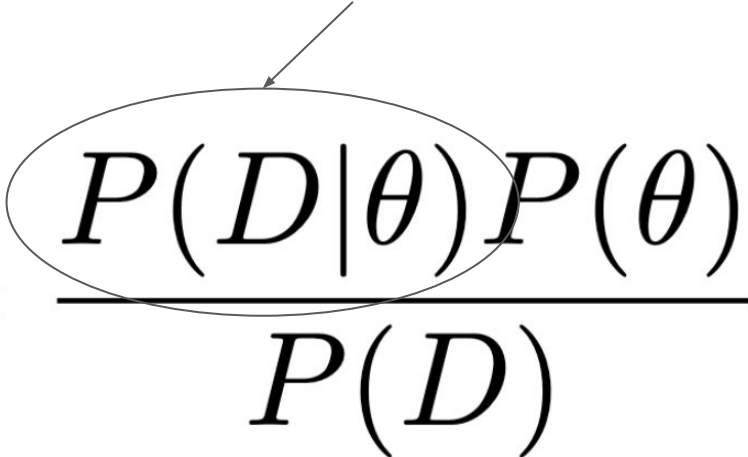
- SBI is an emerging field of statistical inference
 - Also known as likelihood-free inference (LFI)
- Roots in Approximate Bayesian Computation (ABC), but much more efficient
 - Typically uses neural networks
- To do SBI, all you need to be able to do is simulate your data
- SBI provides a tool for doing statistical inference in cases which would otherwise be impossible or require a number of (incorrect) assumptions

Neural Posterior Estimation


$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

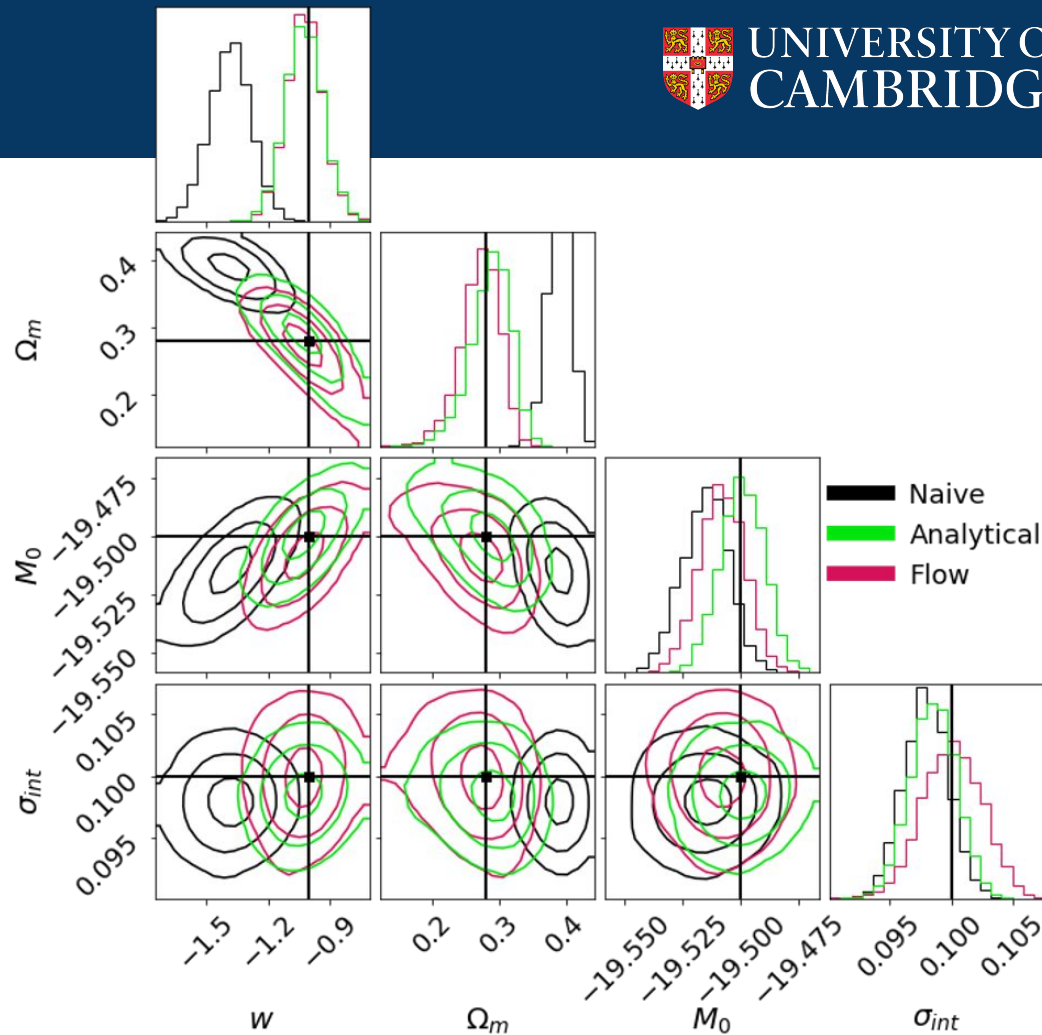
Neural Likelihood Estimation

Neural Posterior Estimation

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$


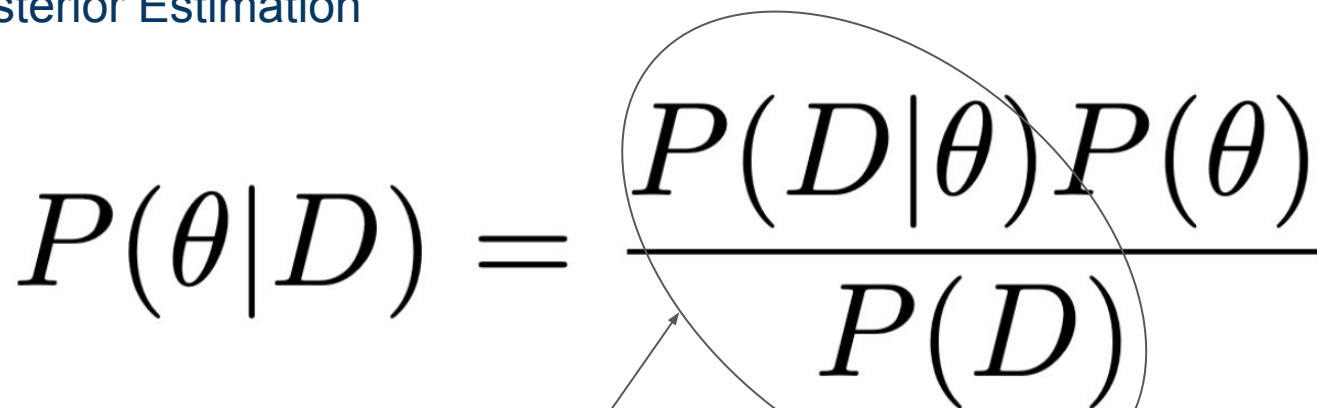
An Example of NLE

- Work in progress by Ben Boyd
- Uses normalising flows to learn likelihood function given selection effects
- Can embed these within current MCMC analyses



Neural Likelihood Estimation

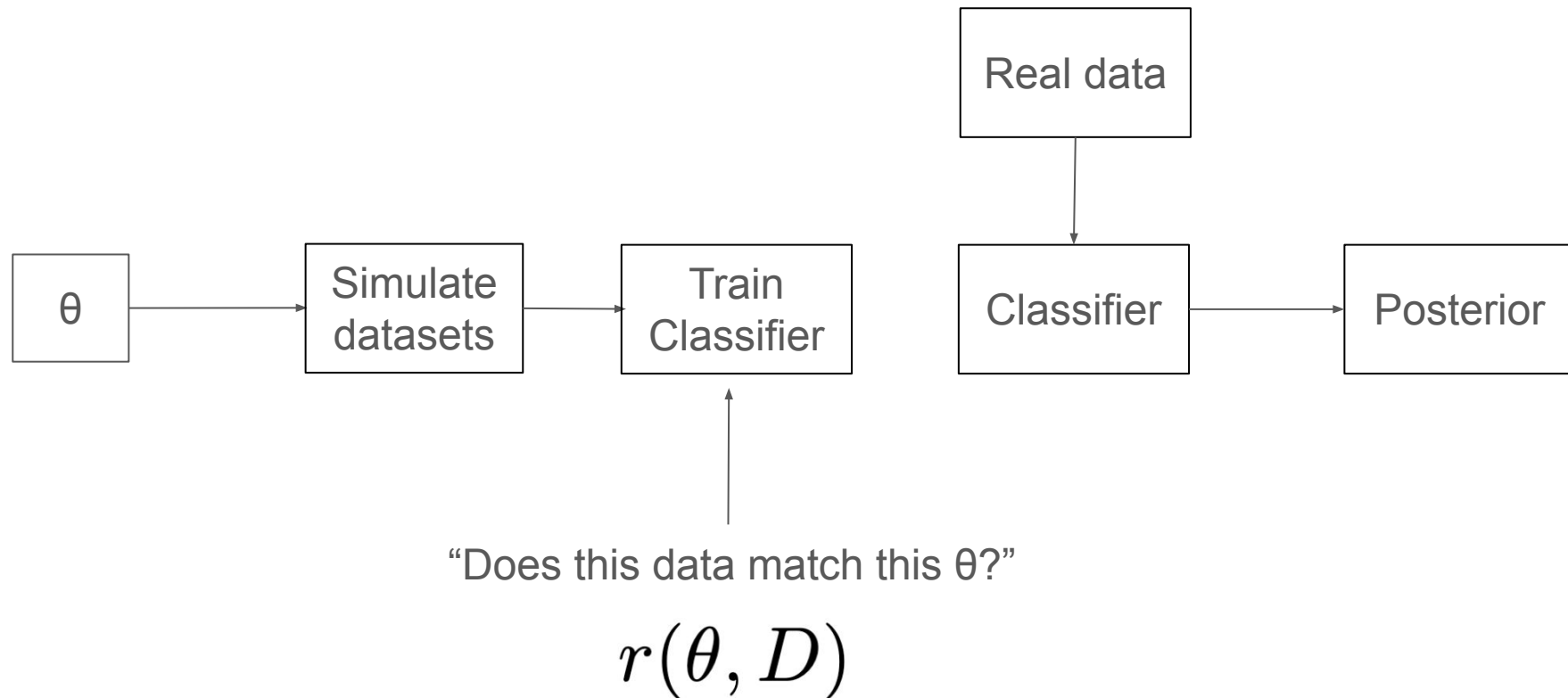
Neural Posterior Estimation

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$


Neural Ratio Estimation

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)} \equiv r(\theta, D)P(\theta)$$

$$r(\theta, D) = \frac{P(D|\theta)}{P(D)} = \frac{P(\theta, D)}{P(\theta)P(D)}$$



Doesn't this rely on your simulations
being a good description of the data?

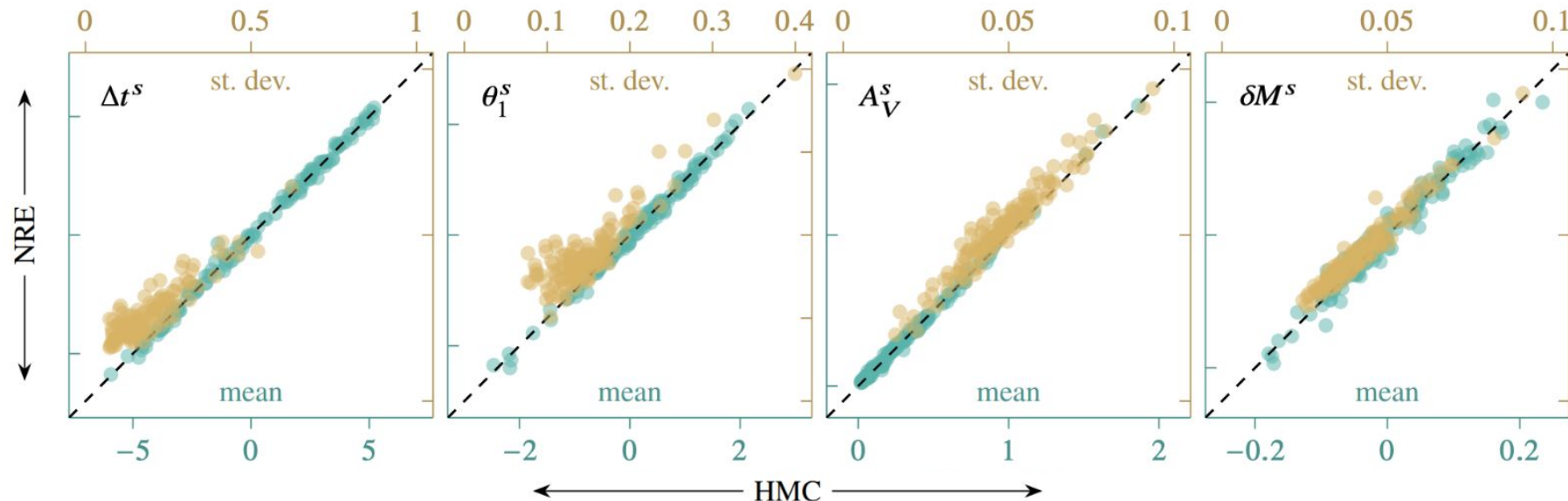
Doesn't this rely on your simulations
being a good description of the data?

Yes...but so do other
approaches

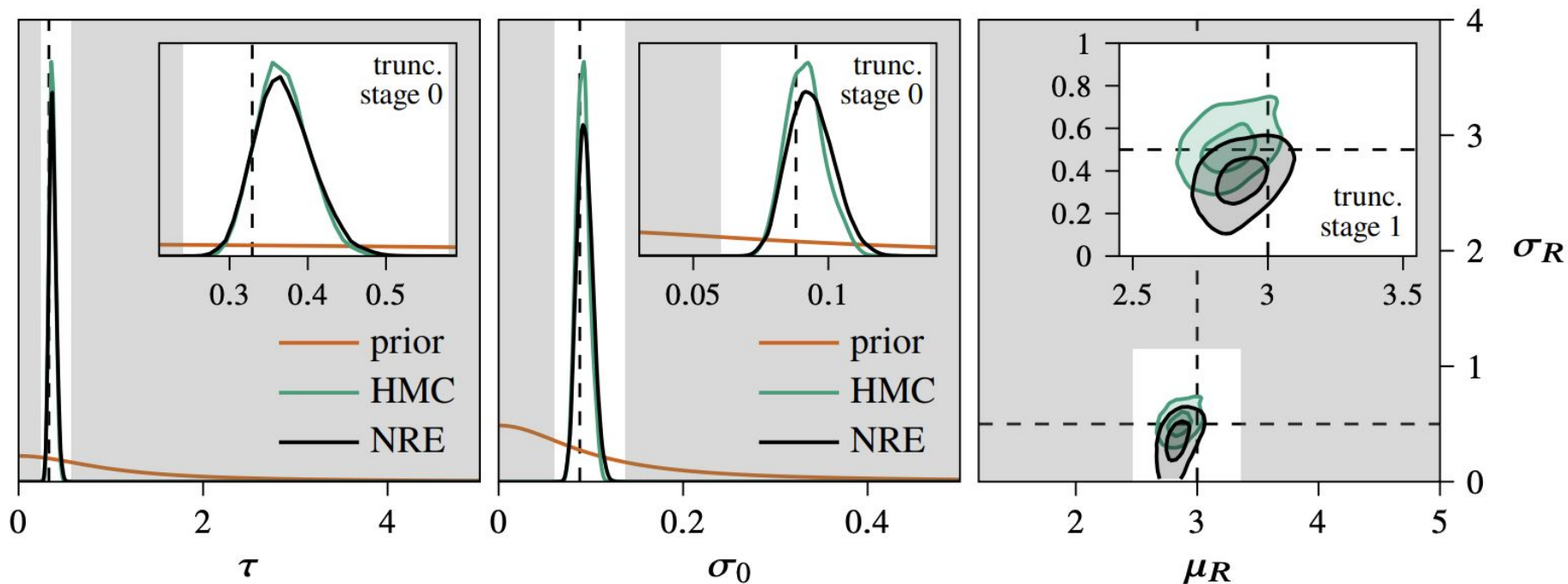
- We can simulate SN samples including a number of complex effects which could not be included in likelihood-based analyses
 - Arbitrary selection effects
 - Photometric redshifts
 - Photometric classification/contamination
- TMNRE provides a path to including these effects within a Bayesian framework

- Karchev, Grayling et al. 2024 recently applied truncated marginal neural ratio estimation (TMNRE) to dust inference of type Ia supernovae
- Light curves simulated from BayeSN model using custom-built PyTorch-based SN simulator *slicsim* (<https://github.com/kosiokarchev/slicsim>)
- This intermediate analysis was designed to allow for a comparison between TMNRE and HMC

SBI for Dust Inference



SBI for Dust Inference



The Future of Supernova Cosmology?

- TMNRE will allow us to incorporate a variety of complex effects within Bayesian analyses of supernova samples
- Allows full end-to-end analysis in single model, cosmology marginalised over the SN model
- Relies on simulations similar to those we already use (mostly)

Conclusions

- 1) BayeSN provides many advantages for inference of SN Ia populations and distances over other approaches
- 2) Through hierarchical inference either side of the mass step, BayeSN supports a mass step driven at least in part by intrinsic differences although dust may also play a role
- 3) TMNRE provides a path towards a fully hierarchical cosmological analysis, and can be a vital tool in the future